

Green synthesis of silver nanoparticles using *Caesalpinia pulcherrima* flower extracts: Evaluating their phytochemicals and antioxidant activity

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Abstract:

The contemporary advancement of nanobiotechnology, where nanoparticles are green synthesised using biological bottom-up approach to overcome the limitations of conventional methods has gained attention in numerous fields, including medicine. This study aimed to green synthesize silver nanoparticles (AgNPs) using five varieties of *Caesalpinia pulcherrima* (CP) (Orange, Yellow, Red, Pink Yellow and Pink) in an environmentally friendly and sustainable approach and determine their antioxidant, photocatalytic and antibacterial properties. The AgNPs were synthesised at optimum temperature at 90 °C for 60 minutes and were characterized using UV-visible spectrum, that showed a plasmon resonance peak between 400-460 nm. Transmission electron microscopy analysis revealed the synthesised CP AgNPs are spherical in shape and 20 nm in size. The biosynthesized AgNPs could be used in treating free radical-mediated diseases and multidrug resistant bacterial infections and minimize environmental pollution by degrading organic dyes to non-toxic intermediate compounds.

1. Introduction

Nanotechnology is a growing field of science and engineering concerned with the study, manipulation, and application of materials with at least one dimension in the nanoscale (1–100 nm) to fabricate materials, systems, and technologies with fundamentally distinct traits and functionalities.¹ It allows the utilization of nanoparticles with unique physical, chemical, optical, and electrical properties in a wide array of fields, placing them at the forefront of technology.² Asghari *et al.* describe nanobiotechnology as a branch of study that investigates the unique physicochemical and biological features of nanoparticles and their applicability in various fields, including health and agriculture (Figure 1).³ Using physical, chemical, and biological techniques, nanoparticles (NPs) are fabricated. Physical and chemical approaches are stereotypical approaches that have drawbacks. (Figure 2). These methods consume high energy, form hazardous by-products, utilize toxic compounds, and produce substandard products with less purity. The limitations are overcome by biological techniques or “green synthesis”.^{4–6}

Green synthesis is the eco-friendly technique of NP synthesis utilizing biological samples such as plants, microbes, and fungi. Plant green synthesis has numerous advantages, including fast, high-yield production, low cost, and non-toxicity.^{7,8} AgNPs synthesis utilizes water,

maintaining a non-toxic, environmentally friendly approach over organic solvents in plant extracts.⁹

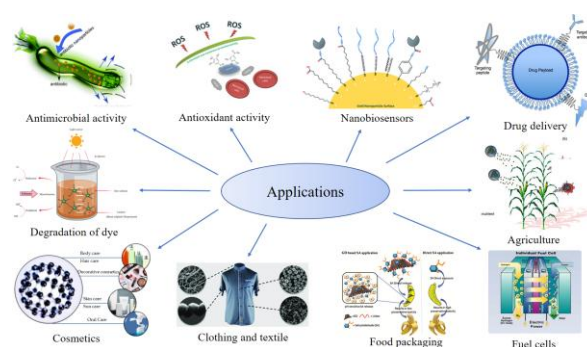


Figure 1. Applications for nanotechnology

Silver nanoparticles (AgNPs) are fabricated in this research due to their unique properties; unlike zinc, gold, and copper, silver has distinctive properties such as thermal and electrical conductivity, chemical stability, and catalytic activity. Due to their size and high surface-to-volume ratios, nanoparticles exhibit different physical and chemical properties than their conventional counterparts, including biological traits, electrical and thermal conductivity, and photocatalytic activity.^{10,11} Furthermore, AgNPs demonstrate surface plasmon resonance (SPR) and are highly resistant to bacteria, rendering them excellent for biological application.¹²

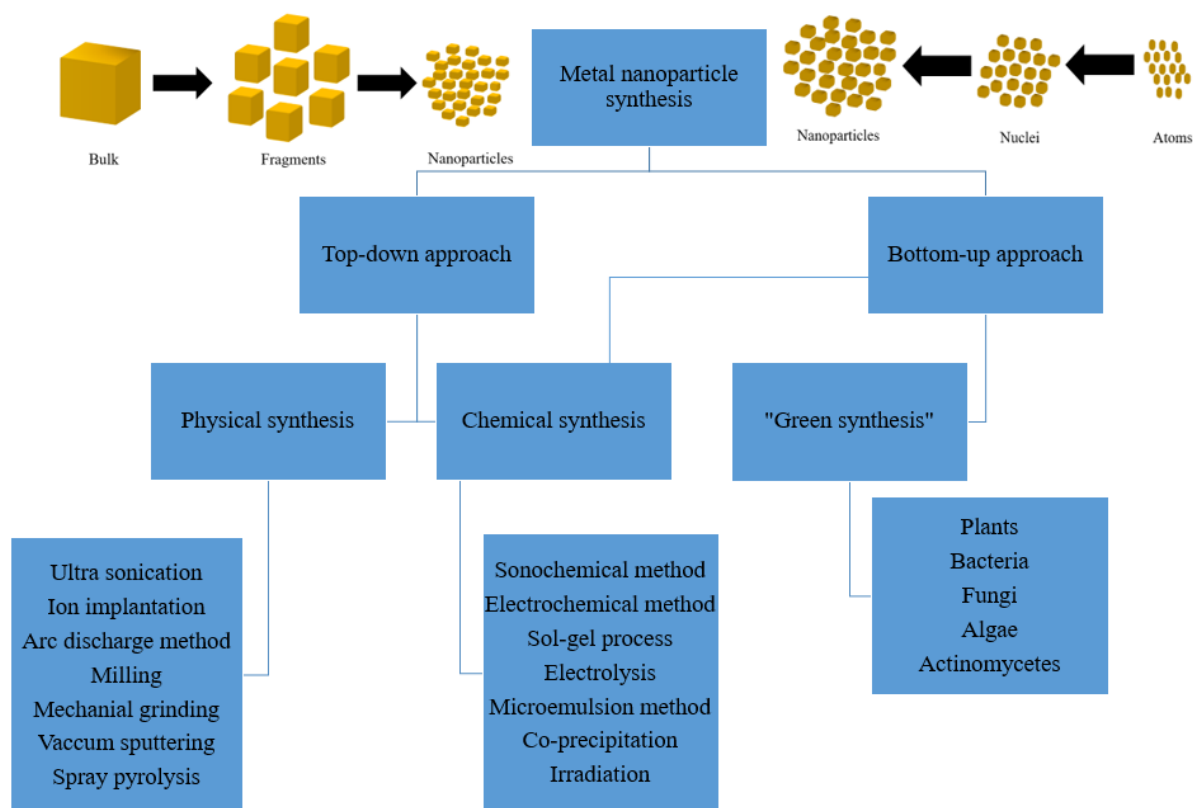


Figure 2. The approaches for nanoparticle synthesis

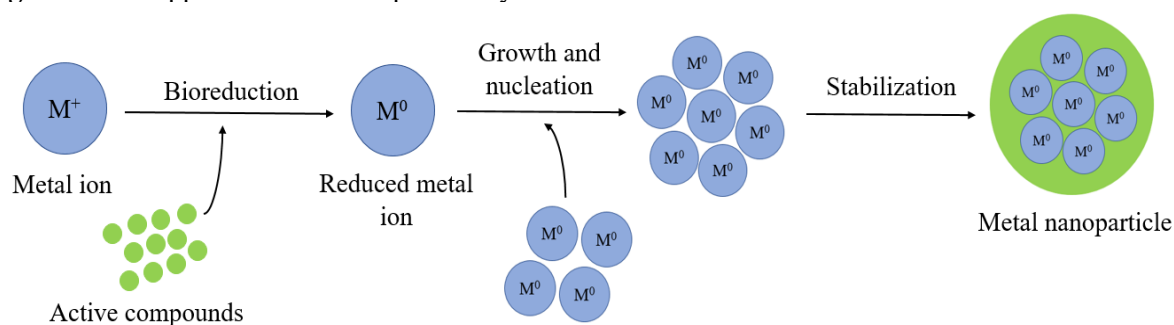


Figure 3. Mechanism of AgNPs green synthesis

In green synthesis of metal nanoparticles (MNP), a metal solution is added to a biological extract and, with the help of active compounds present in the biological sample at different conditions such as concentrations, pH, temperature, pressure, and time, the bio reduction occurs.

During the reduction process, the monovalent or divalent metal ions are reduced to a zero-valent state by the present phytochemicals and other factors including temperature, pH, pressure, and reaction time. This is accompanied by nucleation and growth and

eventually stabilization, where phytochemicals function as capping agents and stabilize the nucleation, forming nanoparticles (Figure 3).^{13,14}

In this research, *Caesalpinia pulcherrima* (CP) flower extracts were used to synthesize AgNPs. CP, also known as peacock flower, is a member of the Fabaceae family of legumes (Figure 4). Traditional medicinal and herbal advantages from various parts of China have been well recognized across the world. It is used for treating fevers, ulcers, asthma, tumors, and skin



Figure 4. *Caesalpinia pulcherrima* of Fabaceae family (Sharma, Goyal and Bhat, 2013)

diseases. They are known for their medicinal properties against malaria, cholera, and bronchitis and have been used due to their anti-inflammatory, antioxidant, analgesic, antidiabetic, and anthelmintic properties. Studies identify that CP is rich in phenols that have high antioxidants and antibacterial properties.^{14–16} Floral waste is produced when *Caesalpinia pulcherrima* is used as an ornamental plant. NP synthesis using floral waste prevents wastage of resources. Khatami et al reported the similar research using hay to synthesize AgNPs.¹⁷

Antioxidants, found naturally in the body or from food, protect against harmful oxidants. When oxidants like ROS, free radicals, and RNS overwhelm antioxidants, oxidative stress occurs, damaging cells and increasing disease risk. Research shows that nanoparticles from antioxidant-rich natural sources can replace toxic synthetic antioxidants like BHT-Q and BHA, which have been linked to health issues such as allergies, DNA damage, and liver cancer, supporting the move toward green synthesis (Figure 5).^{18–20}

The objectives of this study are to green synthesis AgNPs from five different CP flower extracts and assess their antioxidant, photocatalytic, and antibacterial properties. Testing for total flavonoid content, total phenolic content, total antioxidant capacity, 2,2-diphenyl-1-picrylhydrazyl (DPPH) scavenging activity and median inhibition concentration (IC_{50}) tests will be conducted to assess antioxidant activity.

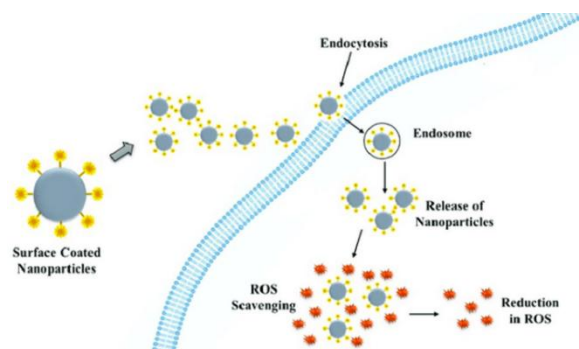


Figure 5. Antioxidant mechanism of nanoparticles (adapted from Bhardwaj *et al.*, 2020)

2. Methodology

All procedures followed good laboratory practice and safety protocols (COSHH and bioCOSHH).

2.1 Sample collection and preparation

Five varieties of *Caesalpinia pulcherrima* (CP) flowers were collected from home gardens in Colombo, Sri Lanka (Figure 6). Flowers were air-dried for 10 days, ground, and 2 g was boiled with 50 mL distilled water for 10 minutes. Extracts were filtered and stored at 4 °C.

2.2 Phytochemical analysis

Conducted as outlined in Table 1.

2.3 Green synthesis of AgNPs

1 mL of extract was mixed with 9 mL of 1 mM $AgNO_3$ and incubated at 90 °C for 60 minutes. Absorbance was measured at 320–520 nm.

2.4 Optimization of AgNPs

Synthesis was optimized at various temperatures and times; absorbance was recorded at 320–520 nm.

2.5 Dilution

Extracts and AgNPs were diluted at a 1:15 ratio and stored at 4 °C.

2.6 TEM Analysis

RF AgNPs were centrifuged, dried at 40 °C, and sent to SLINTEC for TEM analysis.

2.7 TFC, TPC, TAC determination

TFC: $AlCl_3$ colorimetric assay, results in μg QE/100 g. *TPC*: Folin-Ciocalteu assay, results in g GAE/100 g. *TAC*: Phosphomolybdenum assay, results in g AAE/100 g. All assays were in triplicate.

2.8 DPPH scavenging and IC_{50} .

DPPH assay assessed antioxidant activity at 517 nm. IC_{50} was determined using five



Figure 6. Different varieties of *Caesalpinia pulcherrima* flowers used in AgNPs synthesis

Table 1. Procedures for the phytochemical analysis (Roghini and Vijayalakshmi, 2018).

Phytochemical	Procedure	Expected positive results
Tannins	2 mL of 5% FeCl ₃ was added to 1 mL of the extract	Formation of greenish black or dark blue colour
Terpenoids	2 mL of Chloroform and 2 mL of H ₂ SO ₄ was added to 0.5 mL of the extract	Reddish brown colour Formation at the interface
Saponins	2 mL of the extract and 2 mL of distilled water was added to a graduated cylinder and shook well for 15 minutes	Formation of 1cm layer of foam
Carbohydrate	1 mL of Molisch's reagent was added to 2 mL of extract and followed by addition of few drops of concentrated H ₂ SO ₄	Formation of reddish or purple ring
Proteins	Few drops of 0.2% ninhydrin were added to 2 mL of the extract	Formation of mauve or purple colour
Steroids	1 mL of Chloroform was added to 1 mL of the extract, followed by the addition of few drops of H ₂ SO ₄	Formation of brown ring
Anthraquinones	Few drops of 10% NH ₄ OH was added to 1 mL of extract	Pink coloured precipitate formation

Table 2. Phytochemical analysis test results

Phytochemical	OF	YF	RF	PYF	PF	Results
Tannins	√	√	√	√	√	
Terpenoids	√	√	√	√	√	
Saponins	√	√	√	√	√	
Carbohydrate	√	√	√	√	√	
Proteins	×	×	×	×	×	
Steroids	×	×	×	×	×	
Anthraquinone s	×	×	×	×	×	

concentrations (20–100%) and equation (1).

$$\text{Inhibition (\%)} = \left(\frac{A_{\text{control}}}{A_{\text{sample}}} \right) \times 100 \quad (1)$$

3. Results

3.1 Phytochemical Analysis

The colour change indicated the presence of phytochemicals in the extract (Table 2). According to the results obtained the presence of carbohydrates, tannins, saponins and terpenoids were confirmed.

3.2 Green synthesis of AgNPs

The formation of AgNPs was observed initially by the change from pale yellowish colour to reddish-brown colour in all the samples after incubated at 90 °C for 60 minutes. as shown in Figure 7.

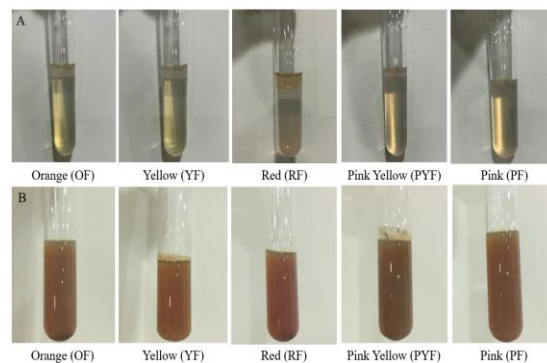


Figure 7. Green synthesis of AgNPs. (A) CP water extracts mixed with silver nitrate before incubation and (B) AgNPs produced after incubation for 60 minutes at 90°C.

3.3 Characterization of AgNPs

The peaks at 420 nm and 440 nm in the UV-Vis spectrum confirmed the formation of AgNPs for all the five varieties (Figure 8).

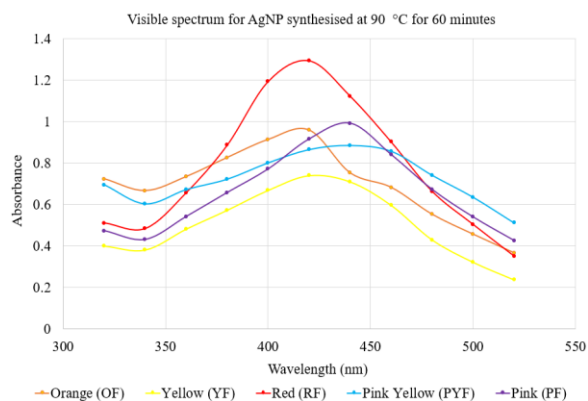


Figure 8. Visible spectrum for AgNPs synthesised at 90°C for 60 minutes.

Transmission Electron Microscope analysis reveals the AgNPs to be spherical and around 40 nm in size (Figure 9).

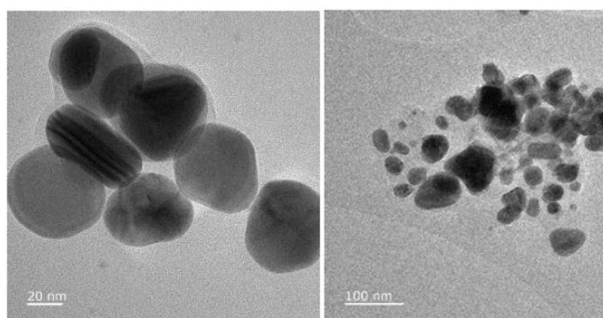


Figure 9. Transmission Electron Microscope images of RF AgNPs

3.4 Determination of antioxidant activity

3.4.1 Total flavonoid content (TFC)

TFC analysis indicated that AgNPs contained has higher TFC than the extracts and RF AgNPs has the highest TFC (Figure 10). The significance difference between extracts and AgNPs was compared by One-way ANOVA.

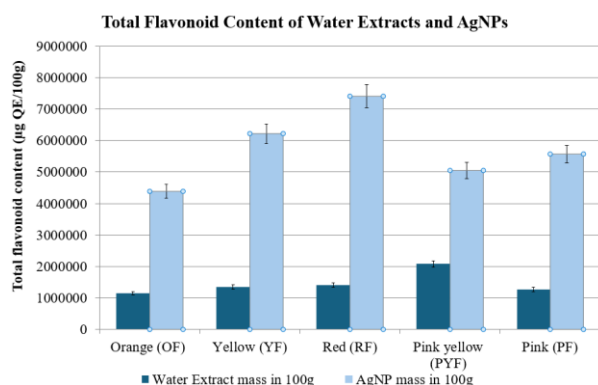


Figure 10. Total flavonoid content in water extracts and AgNPs

3.4.2 Total phenolic content (TPC)

TPC analysis indicated that AgNPs contained have higher TPC than the extracts on average and OF, YF and PYF AgNPs have the highest TPC (Figure 11). The significance difference between extracts and AgNPs was compared by One-way ANOVA.

3.4.3 Total antioxidant capacity (TAC)

TAC analysis indicated that AgNPs contained has higher TAC than the extracts and PYF AgNPs have the highest TAC (Figure 12). The significance difference between extracts and AgNPs was compared by One-way ANOVA.

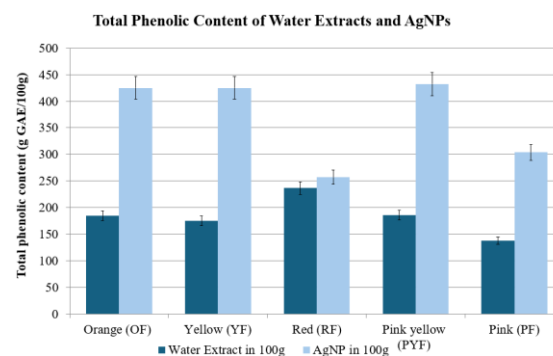


Figure 11. Total phenolic content in water extracts and AgNPs

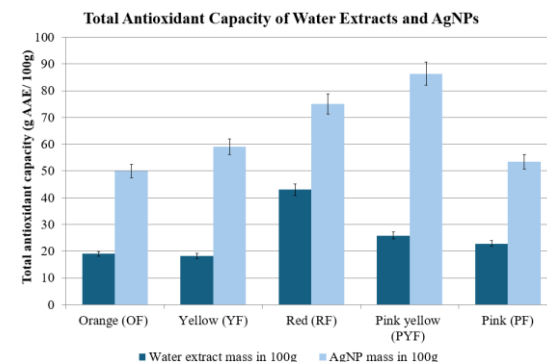


Figure 12. Total antioxidant capacity in water extracts and AgNPs

3.4.4 Determination of DPPH scavenging activity

AgNPs when compared to the extracts showed similar levels of % DPPH scavenging activity excluding RF and PYF (Figure 13).

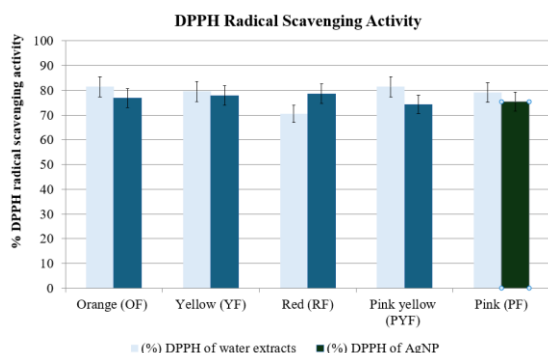


Figure 13. % DPPH scavenging activity of water extracts and AgNPs

3.4.5 Determination of median inhibitory concentration (IC_{50}) of DPPH

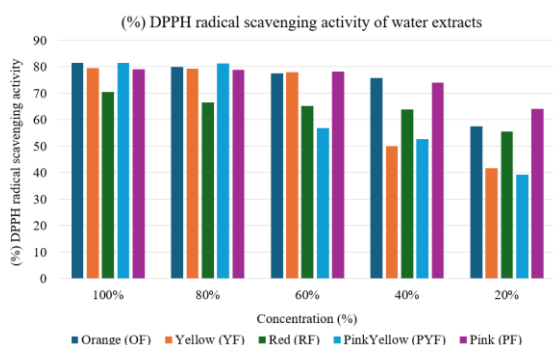


Figure 14. IC_{50} of water extracts

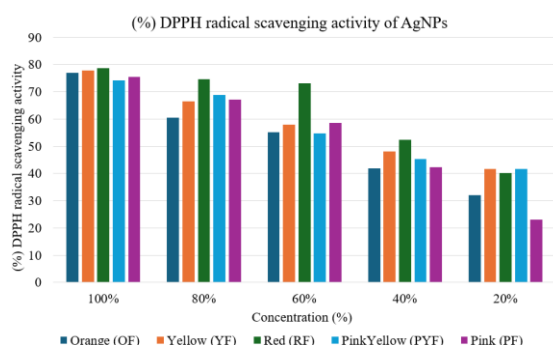


Figure 15. IC_{50} of AgNPs

Table 3. IC_{50} of water extracts and AgNPs

Sample	IC_{50} of water extracts (%)	IC_{50} of AgNPs (%)
OF	0.34	0.54
YF	0.30	0.42
RF	0.28	0.32
PYF	0.38	0.44
PF	0.84	0.55

The IC_{50} of PF AgNPs was lower than its extracts, which indicates a higher antioxidant capacity (Table 3) (Figures 14–15).

4. Conclusion

This study successfully demonstrated the green synthesis of silver nanoparticles (AgNPs) using five varieties of *Caesalpinia pulcherrima* flowers, offering an eco-friendly and sustainable alternative to conventional methods. The synthesized AgNPs exhibited strong antioxidant activities, with superior performance compared to crude water extracts. Characterization confirmed their nanoscale size, spherical shape, and semiconductor properties. These findings highlight the potential of CP-based AgNPs in biomedical and environmental applications, particularly for treating oxidative stress-related diseases. Therefore, a detailed understanding of the properties of each variety is facilitated by these research findings, allowing a better alleviation of the dilemma faced by humankind and the environment.

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Research Ethics / Declarations

Authors declare no conflict of interest.

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